

A Study on Sonification Method of Simulator-Based Ski Training for People with Visual Impairment

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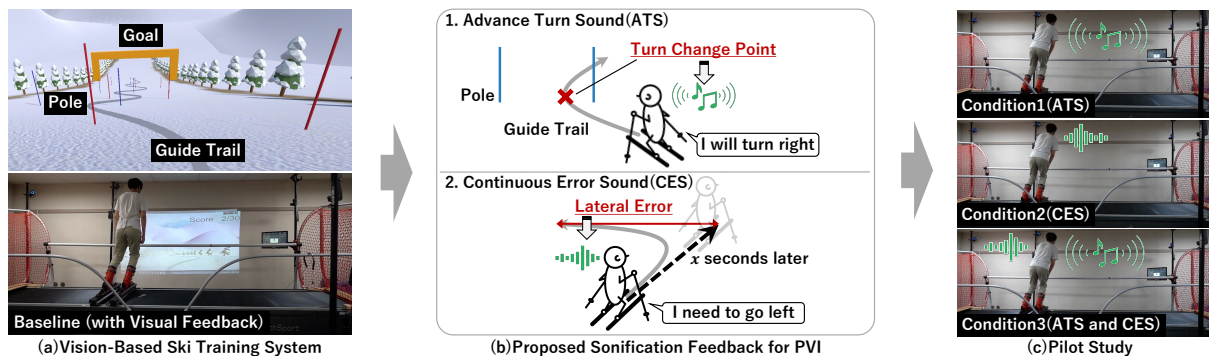


Figure 1: Overview of the Approach

ABSTRACT

People with visual impairment (PVI) are eager to push their limits in extreme sports such as alpine skiing. However, training skiing is very difficult as it always requires assistance from an experienced guide. This paper explores sonification-based methods that enable PVI to train skiing using a simulator, which will allow them to train without a guide. Two types of sonification feedback for PVI are proposed and studied in our experiment. The results suggest that users without any visual information can also pass through over 80% of the poles compared to with visual information on average.

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1 INTRODUCTION

Indoor ski simulators (Figure 1-a) allow skiers to practice skiing all year round without any environmental restrictions. However, people with visual impairment (PVI) can not use the simulators, as they employ only visual information as feedback. In actual blind

skiing, a sighted guide skis in front of or behind PVI to inform them of the direction and timing of turns with their voice. Therefore, sighted guides are necessary for ski training for PVI, however, the number of guides is far from sufficient. To enable solo training for PVI, we explore two types of sonification feedback (Figure 1-b), an Advance Turn Sound (ATS) which inform the turn timing, and a Continuous Error Sound (CES) which feedback a lateral error, based on an interview of visually impaired (VI) skiers and their guides. In a pilot study, we study the effectiveness of the two proposed methods and their mixed version on a ski simulator to investigate whether the proposed feedback is useful for PVI to use the simulator alone. Finally, the results are compared within each sonification condition and compared to a baseline with visual information.

2 RELATED WORK

There have been studies on improving the efficiency of ski training using ski simulators [Matsumoto et al. 2022; Wu et al. 2019] for sighted people, but not for PVI. On the other hand, various studies have been conducted to assist PVI in following a predetermined path with sonification feedback. Parseihian *et al.* [Parseihian et al. 2018] showed that feedback considering both the current and future position of blind people is effective in a path-following task. However, there is no research on effective feedback for tasks such as skiing, in which a user continuously makes quick turns using the whole body. Therefore, in this study, we designed the sonification feedback suitable for ski training for PVI.

3 IMPLEMENTATION

Two types of sonification methods are designed based on an initial interview with VI skiers and their guides.

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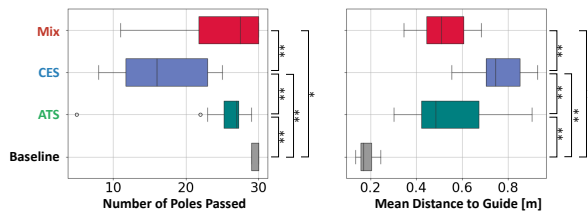


Figure 2: Results of the pilot study. The brackets indicate the significant differences: *($p < .05$), **($p < .01$).

Advance Turn Sound (ATS). This sound is designed to signal the timing, the intensity, and the direction of the next turn (Figure 1-b1), which is emitted just before the turn from the direction of the next turn by stereo audio. To signal the intensity of the turn, we use a low tone which represents a large turn, and a high tone for a small turn. This is based on the actual guides' call which is called out just before the turn to convey to PVI the intensity of a turn by controlling the pitch of their voice.

Continuous Error Sound (CES). This sound corresponds to another call from the guides, which uses the volume and length of a call to inform the user how far and in what direction the user needs to move from the current position (Figure 1-b2). Based on the previous study by Parseihian *et al.* [Parseihian *et al.* 2018], we raise the pitch of the sound when the lateral error between the user's future position and the guide's position is large. This is aimed at conveying how far the user needs to move in a horizontal direction. Also, to indicate the direction to move, the sound is emitted from the direction in which the guide is at using stereo audio.

Additional Successful Sound. A successful sound is emitted when the user passes the poles, which aims to provide post-event feedback for the user as well as to increase their motivation.

4 PILOT STUDY

We conducted a pilot study with six sighted subjects (2 females, 4 males, Mean age=23.7, SD=1.80) who had previously used a ski simulator. For the experiment, we created a virtual slope with 30 poles and placed a trajectory of a guide as a gray line on the course [Matsumoto *et al.* 2022]. The users were asked to pass through the poles and follow the trail of the guide. The purposes of this experiment are to compare the baseline (B), which is skied with only visual information, with our proposed sonification methods, and to investigate the effectiveness of the proposed methods. For this purpose, we prepared three conditions: C1 (ATS), C2 (CES), and C3 (ATS and CES). In this experiment, successful sounds were emitted in C1, C2, and C3. First, participants conducted a training session with the projector on for 2 minutes, and then they performed a baseline run. Next, the projector was turned off (Figure 1-c), and a 2 minutes training session of each condition (C1–C3) was conducted. After the training, the users skied two courses in each condition in random order. We used randomly generated courses to avoid memorizing a course. Finally, we evaluated the user's runs using the number of passed poles and the mean distance between the virtual guide and the user's position.

5 RESULTS AND DISCUSSION

Figure 2 shows the results for each condition. The results show that in C1 and C3, users were able to pass through over 82% of the poles on average compared to the baseline. Mann-Whitney's U test is applied to compare each condition and significant differences are found between other conditions and baseline (B and C1: $p < .01$, B and C2: $p < .01$, B and C3: $p < .05$), also between C1 and C2 and between C2 and C3 (C1 and C2: $p < .01$, C2 and C3: $p < .01$). These results indicate that ATS is more effective than CES in virtual alpine skiing, where quick reactions are required.

For C1 and C3, the results are more controversial. In terms of passed poles, the numbers vary greatly in C3 while for the mean distance to guide, C1 is more unstable. Although there is no significant difference between C1 and C3, we can see a tendency that combining ATS with CES can improve the distance but will affect the timing of a turn. More insights are found in the interview. Although CES (C2) performs the worst, we received many comments saying it was useful when users were far off the trajectory. In addition, we obtained many comments that C3 was confusing because there were many situations where multiple sounds were superimposed, which might explain the above quantitative results.

Based on the above results, we are considering developing a condition to emit CES when users are far off the guide's trail and emit ATS when they are on the trail to solve the problem of information overload. In addition, the current pilot study was conducted with eyes-opened sighted people. However, the way PVI perceives the sonification feedback might be different from sighted people. Furthermore, the timing parameter of both sounds is not studied in the pilot experiment, which might have a significant impact on users' performance. The interview with guides shows that the optimal timing for calling out the call which corresponds to the ATS varies from skier to skier. Therefore, we will investigate the relationship between the guiding voice of a professional guide and VI skier's performance to discover the optimal timing of sound for them.

6 CONCLUSION

This paper explores the sonification-based feedback that enables PVI to train skiing using a ski simulator. In a pilot study, in conditions where only ATS was emitted, users could pass through about 82% of the poles compared to a baseline on average, demonstrating the effectiveness of the ATS. Based on the results of the pilot test and interviews, we plan to implement sonification feedback for a more accurate trail following and conduct experiments with PVI.

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REFERENCES

- Takashi Matsumoto *et al.* 2022. Skiing, Fast and Slow: Evaluation of Time Distortion for VR Ski Training. In *Augmented Humans 2022* (Kashiwa, Chiba, Japan) (AHs 2022). Association for Computing Machinery, New York, NY, USA, 142–151. <https://doi.org/10.1145/3519391.3519402>
- Gaëtan Parseihian *et al.* 2018. Exploration of Sonification Strategies for Guidance in a Blind Driving Game. In *Music Technology with Swing*. Springer International Publishing, Cham, 413–428.
- Erwin Wu *et al.* 2019. How to VizSki: Visualizing Captured Skier Motion in a VR Ski Training Simulator. In *The 17th International Conference on Virtual-Reality Continuum and Its Applications in Industry* (Brisbane, QLD, Australia) (VRCAI '19). Association for Computing Machinery, New York, NY, USA, Article 5, 9 pages. <https://doi.org/10.1145/3359997.3365698>